



Plan Your Algorithm

Plan it, write it, then make it better

Name:

Date:

★ I can plan and write my own algorithm, then improve it.

You are the coder

Plan an algorithm: think of the steps, write them in order, then check by doing it in your head. Two jobs to plan — a pattern and a real task — then make one better.

Pattern boxes 1 to 6

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1 Plan a pattern

Plan a pattern of 6 shapes using ● and ■. Write your steps (box 1 → draw ?, box 2 → draw ?, ...), then draw your pattern in the boxes above.

2 Plan a real job

Pick a job you know — pack your backpack, make a snack, or get ready for school. Write the steps in order (4 to 6 steps), one per line.

3 Make it better

Read your job algorithm. Add ONE step, or fix the order, to make it better. Write your change and tell why it helps.

Big idea

You can read, write, follow, and plan algorithms — and make them better. That is what coders do every day. You are a coder!



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Quick facilitation notes & answer key

What this worksheet teaches

The unit finale: students plan and write a full set of steps for a pattern and for a real task, then improve one by adding a step or fixing the order. Read, write, and improve a plan — all on one page. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: cut the steps into cards so students can physically slide them into order before writing.

How to lead it (about 20–25 minutes)

1. Show them (you do). Plan a 6-shape pattern by writing what goes in each box, then drawing it.
2. Do one together. Start a task plan together (e.g. pack your backpack) — steps in order.
3. Let them try. Students finish their task plan, then improve it by adding a step or fixing the order.

Answer key (these are the verified correct answers)

Exercise 1 — open; a sample is circle, square, square, circle, square, square. Accept any pattern where the steps match the drawing.

Exercise 2 — open; a sample (pack your backpack): open the backpack, put in your books, put in your lunch, zip it up, put it on. Accept any 4–6 ordered steps.

Exercise 3 — open; a sample improvement: add "check your homework is in" before "zip it up". Accept any added step or reorder with a reason.

If students get stuck — and ways to adjust

Watch for: a drawing that does not match the written steps. Check each step against each box.

Make it easier: let them reuse a task from an earlier sheet.

Make it harder: ask them to write a plan with one flexible pair of steps.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students put step-by-step instructions in the right order to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a set of steps, change or reorder one, and explain what the change did.

You'll know it worked when

The child writes a working plan for both the pattern and the task and improves one with a reason.